ANSWER KEY & MARKING SCHEME SET 1

Chapter 4: **Changes Around Us: Physical and Chemical** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** a reversible chemical change	Hydrated and anhydrous forms.
2	**(c)** formation of a new substance	New substance.
3	**(b)** store in dry place or oil them	Keeps moisture away.
4	**(a)** copper sulphate	Copper deposits.
5	**(d)** lime water test	Test gas.
6	**(c)** apply oil and keep dry	Oiling.
7	**(c)** carbon dioxide	Lime water test.
8	**(d)** unburnt carbon, a chemical change	New substance.
9	**(b)** to prevent rusting and reduce friction	Both reasons.
10	**(b)** conservation of mass	Mass is conserved.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Iron combines with oxygen and water (1); Rust has extra mass (1).	1+1
12	Burning forms ash and gases (1); Tearing only changes size and shape (1).	1+1
13	Takes a long time (1); Small changes each day (1).	1+1
14	No new substance in melting (1); Magnesium oxide forms (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	A: no rust (no moisture) (1); B: no rust (no air) (1); C: rusts (air and water) (1).	1+1+1
16	Salt and sand do not react (1); Iron and sulphur react (1); New compound formed (1).	1+1+1
17	Burning: chemical (1); Light, sound, heat (1); Smoke and gases (1).	1+1+1
	OR: Chemical (1); New substance slaked lime (1); Heat released (1).	1+1+1
18	Burning candle (1); Melting is physical (1); Burning vapour is chemical (1).	1+1+1
	OR: Combines with iron (1); Combines with fuel (1); Oxides form (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Set-ups: dry, no air, tap water, salt water (1); Same nails and period (1); Observe daily (1); Compare rust amounts (1); Salt water most, oiled none (1).	1+1+1+1+1
	OR: Burn a candle in a closed jar (1); It goes out (1); Add lime water (1); It turns milky (1); Conclude (1).	1+1+1+1+1
20	LPG burns: chemical (1); Milk warms: physical (1); Milk boils: physical (1); Milk burns at bottom: chemical (1); Black deposit is carbon (1).	1+1+1+1+1
	OR: Washing dissolves dirt: physical (1); Dry cleaning solvent (1); Silver polish: chemical (1); Rust removal by acid: chemical (1); Choose method (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 4: **Changes Around Us: Physical and Chemical** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** a chemical change	Energy released.
2	**(c)** silver reacts with sulphur in air	Alloy resists.
3	**(c)** a new black solid not attracted by a magnet	New compound.
4	**(a)** melting of ice	Heat taken in.
5	**(b)** to resist rusting	Longer life.
6	**(d)** rust is brittle and flakes off	Weakens.
7	**(d)** burning of LPG	Combustion.
8	**(c)** new products carbon dioxide and water	Combustion.
9	**(d)** proteins change irreversibly, so it is chemical	New substances.
10	**(c)** heat is absorbed in dissolving	Endothermic.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Boiling removes dissolved air (1); So only water, not air, is tested (1).	1+1
12	Higher temperature (1); Steam speeds the change (1).	1+1
13	Water of crystallisation lost can be regained (1); Ash cannot become paper (1).	1+1
14	It does not relight a glowing splinter (1); It turns lime water milky (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Warm place speeds (1); Cold slows (1); New substances formed (1).	1+1+1
16	Both are oxidation (1); Burning is fast, rusting slow (1); Burning gives heat and light (1).	1+1+1
17	Chlorophyll breaks down (1); Other pigments visible (1); Chemical change (1).	1+1+1
	OR: Physical (1); Same substance (1); Energy used for change of state (1).	1+1+1
18	Reacts with chalky carbonate (1); Releases carbon dioxide (1); Chemical change (1).	1+1+1
	OR: Cooking (1); Burning (1); New substances formed (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Dissolving: physical (1); Crystallisation: physical (1); Heating hydrated crystals: chemical (1); Adding water: reversal (1); Differences summarised (1).	1+1+1+1+1
	OR: Coin becomes shiny or green (1); Lemon acid reacts (1); New substances (1); Chemical change (1); Safe handling (1).	1+1+1+1+1
20	Bridges and rails damaged (1); Cost of repairs (1); Paint and galvanise (1); Use stainless steel (1); Regular maintenance (1).	1+1+1+1+1
	OR: Shaping: physical (1); Drying: physical (1); Firing in kiln: chemical (1); Hard pot formed (1); Cannot return to clay (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 4: **Changes Around Us: Physical and Chemical** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** gas bubbles and new substances form	Yeast action.
2	**(c)** a chemical change	New substances such as carbon form.
3	**(c)** a chemical reaction releases heat	Slaking.
4	**(d)** burning of a candle	Mix.
5	**(a)** the batter rises and changes colour	Gas and colour change.
6	**(b)** slaked lime, a new substance, forms	New product.
7	**(c)** reaction with air oxygen is slowed	Isolation from air.
8	**(c)** dissolving salt in water	Only dissolving is physical.
9	**(d)** salt dissolving in water	No new substance.
10	**(b)** acidic or salty water	Electrolyte.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Surface becomes iron oxide (1); Dull flaky layer forms (1).	1+1
12	Yeast produces gas (1); Chemical change from fermentation (1).	1+1
13	Forms silver sulphide (1); Black coating (1).	1+1
14	New substances like carbon form (1); Chemical change (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Zinc coating (1); Alloy with chromium (1); Resist corrosion (1).	1+1+1
16	Crystals contain water (1); Water is released as vapour (1); Condenses (1).	1+1+1
17	Add water to quicklime (1); Hissing and heat (1); Touch the beaker carefully / use thermometer (1).	1+1+1
	OR: Rubber contains sulphur (1); Forms silver sulphide (1); Chemical change (1).	1+1+1
18	Milk to curd: chemical (1); Churning: physical (1); Butter separated (1).	1+1+1
	OR: Glucose reacts with oxygen (1); Energy released (1); Carbon dioxide formed (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Heating sugar chars it (1); Heating copper turns black (1); Heating copper sulphate gives white powder (1); Heating ice only melts it (1); Depends on substance (1).	1+1+1+1+1
	OR: Food spoils faster in heat (1); Cooking faster at high flame (1); Fridge slows (1); Rusting faster in warm moist air (1); Higher temperature speeds reactions (1).	1+1+1+1+1
20	Microbes act on batter (1); Gas bubbles (1); Dough rises (1); Curd forms (1); New substances (1).	1+1+1+1+1
	OR: List steps for each dish (1); Mark physical steps (1); Mark chemical steps (1); Table of results (1); Conclusion (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.